

CD_DA_02

May 2, 2025

0.1 Import Libraries and Load Dataset

```
[9]: import pandas as pd

# Load the dataset (use your path)
df = pd.read_csv("sample_amazon_reviews.csv")

# Show a few rows
df
```

```
[9]:
```

	review_id	product_title	customer_id \
0	7e514167-4831-4706-ba70-b768614cf4cb	Laptop Stand	25387
1	8b29c48d-f66a-4ec5-beb1-69de8c21347e	Laptop Stand	11094
2	729b5206-508f-40c6-8ea6-ff38eab87b98	Smart Watch	29279
3	b8d4ed2c-835b-41de-93e2-1ff86af8f394	Laptop Stand	97387
4	6981597c-9002-46d7-b033-255d9c9b31cc	Fitness Tracker	63879
..	...	...	...
995	d41c6baa-da21-48c2-9f51-14773e67fece	Wireless Earbuds	74653
996	5da91a7b-e19a-4d7a-b557-8a7188e297c5	Fitness Tracker	94055
997	69104d8e-cc22-4a89-90d5-f708f68c17ba	Wireless Earbuds	81546
998	576d7f2a-531f-4fa4-a1b6-7ce003e36a0b	Fitness Tracker	89981
999	39a1f615-fa75-46e4-ba70-3bb32cfda385	Smart Watch	71031

	review_body	star_rating \
0	Space health authority foot respond news try e...	5
1	Environment them hear person chance coach them...	4
2	Throw we something itself safe serve.	5
3	Interesting those cost offer cut eye she hand.	5
4	Under hit suffer standard last other skin baby...	4
..	...	...
995	Well against old edge they rule issue any syst...	5
996	How evening leg clearly opportunity just read ...	1
997	Might kind size trouble raise economy where dr...	2
998	Buy professional that that example at at in pe...	4
999	Surface door particular year street red mornin...	3

```
review_date
0    2025-03-01
```

```

1    2025-04-17
2    2025-01-28
3    2025-02-22
4    2025-02-09
..    ...
995  2024-08-25
996  2025-02-19
997  2025-04-09
998  2025-03-22
999  2024-06-27

```

[1000 rows x 6 columns]

0.2 Data Preprocessing

```

[7]: # Drop rows with missing values
df = df.dropna(subset=["review_body", "product_title", "star_rating"])
df.head()

```

```

[7]:
      review_id  product_title  customer_id \
0  7e514167-4831-4706-ba70-b768614cf4cb  Laptop Stand      25387
1  8b29c48d-f66a-4ec5-beb1-69de8c21347e  Laptop Stand      11094
2  729b5206-508f-40c6-8ea6-ff38eab87b98    Smart Watch      29279
3  b8d4ed2c-835b-41de-93e2-1ff86af8f394  Laptop Stand      97387
4  6981597c-9002-46d7-b033-255d9c9b31cc  Fitness Tracker      63879

      review_body  star_rating  review_date
0  Space health authority foot respond news try e...      5  2025-03-01
1  Environment them hear person chance coach them...      4  2025-04-17
2          Throw we something itself safe serve.      5  2025-01-28
3    Interesting those cost offer cut eye she hand.      5  2025-02-22
4  Under hit suffer standard last other skin baby...      4  2025-02-09

```

0.3 Encode Text and Categorical Data

```

[11]: from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.preprocessing import OneHotEncoder
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.pipeline import FeatureUnion
from sklearn.compose import ColumnTransformer
from sklearn.metrics import classification_report, accuracy_score

# Separate features and target
X = df[["review_body", "product_title"]]
y = df["star_rating"]

```

```

# Define transformers
text_transformer = TfidfVectorizer(max_features=300)
cat_transformer = OneHotEncoder(handle_unknown='ignore')

# Combine them using ColumnTransformer
preprocessor = ColumnTransformer(
    transformers=[
        ("text", text_transformer, "review_body"),
        ("cat", cat_transformer, ["product_title"])
    ]
)

```

0.4 Split the Data (Train/Test)

```

[13]: X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,
    ↪ random_state=42)

```

0.5 Create and Train the Model

```

[15]: from sklearn.pipeline import Pipeline

# Create a pipeline with preprocessing + model
pipeline = Pipeline(steps=[
    ('preprocessor', preprocessor),
    ('classifier', RandomForestClassifier())
])

# Train the model
pipeline.fit(X_train, y_train)

```

```

[15]: Pipeline(steps=[('preprocessor',
    ColumnTransformer(transformers=[('text',
    TfidfVectorizer(max_features=300),
    OneHotEncoder(handle_unknown='ignore'),
    ('classifier', RandomForestClassifier())])
    , ('review_body'),
    ('cat',
    ['product_title'])])),
    ('classifier', RandomForestClassifier())])

```

0.6 Evaluate the Model

```

[17]: # Predict on test set
y_pred = pipeline.predict(X_test)

# Check accuracy and report

```

```
print("Accuracy:", accuracy_score(y_test, y_pred))
print("Classification Report:\n", classification_report(y_test, y_pred))
```

Accuracy: 0.19

Classification Report:

	precision	recall	f1-score	support
1	0.26	0.43	0.33	37
2	0.13	0.21	0.16	38
3	0.00	0.00	0.00	40
4	0.30	0.15	0.20	41
5	0.17	0.18	0.18	44
accuracy			0.19	200
macro avg	0.17	0.19	0.17	200
weighted avg	0.17	0.19	0.17	200

0.7 Predict on New Data

```
[ ]: sample = pd.DataFrame({
    "review_body": ["Amazing sound quality and battery life"],
    "product_title": ["Bluetooth Speaker"]
})

predicted_rating = pipeline.predict(sample)
print("Predicted Rating:", predicted_rating)
```

0.8 Results

- The model achieved an accuracy of around (your accuracy here)%
- The most influential feature was the **review text**
- This classification model can be extended to real-world review analysis systems

```
[ ]:
```